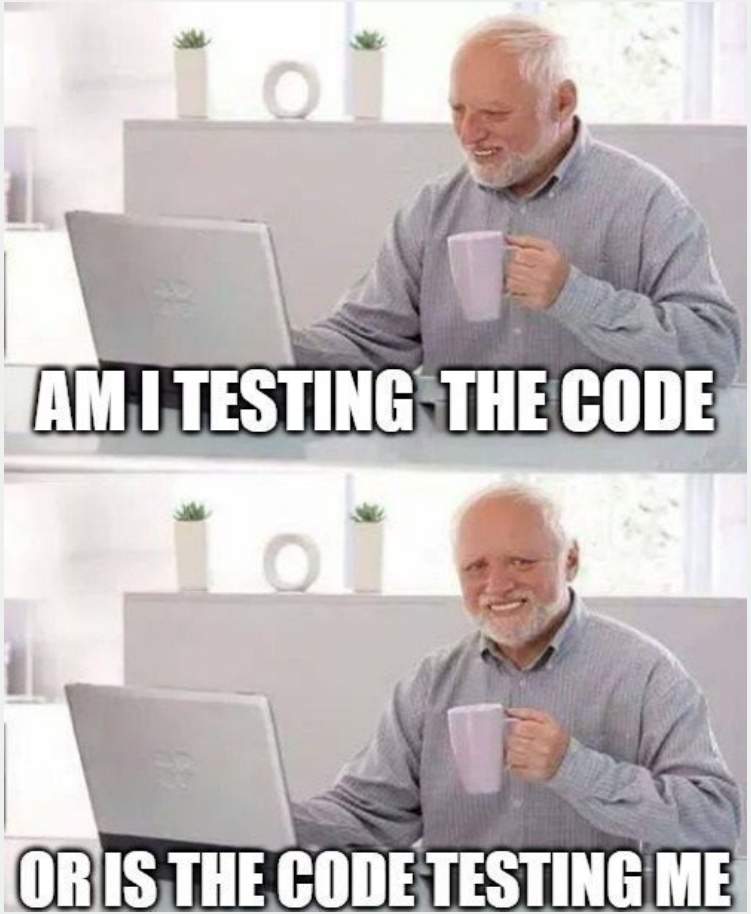
LAB #5

Adv. CiC
Spring 2026

Agenda

1. Housekeeping
2. Data Cleaning
3. Testing
4. Lab 5 Exercise



Housekeeping

Due this past week

- Reading 5
- Lab 4
- Project, pt. 2

*If you use AI for your responses or labs, ensure your work is **still correct and high quality**.*

Upcoming

- Project pt. 3 - **2/26 @ 3pm**
- Reading 6 - **2/26 @ 3pm**
- Lab 5 - **2/27 @ 1pm**

Announcements

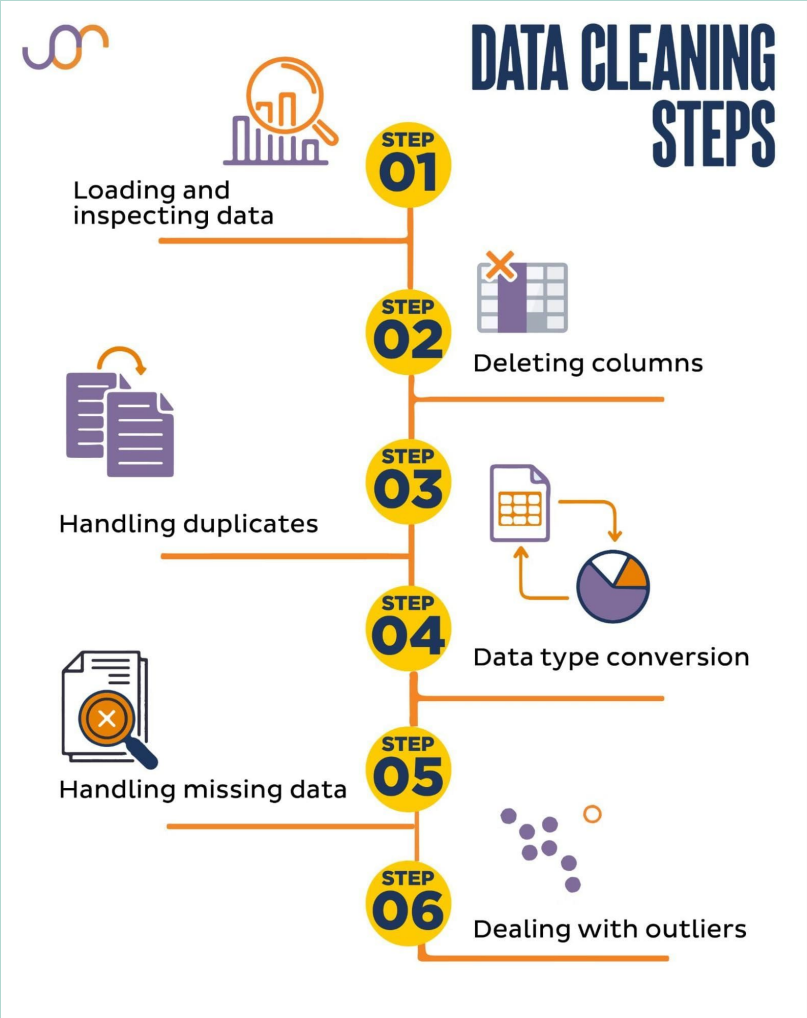
1. Assignment expectations
   a. RRs & Labs
2. Grading questions
3. Survey feedback
4. GitHub repo set-up

Questions
(on Lectures, Labs, or Readings)?

Data Cleaning

What is data cleaning?

the process of identifying and fixing errors, inconsistencies, or missing values in raw data so it can be accurately analyzed or used in a program


DATA CLEANING STEPS
STEP 01
Loading and inspecting data
STEP 02
Deleting columns
STEP 03
Handling duplicates
STEP 04
Data type conversion
STEP 05
Handling missing data
STEP 06
Dealing with outliers

DATA CLEANING CHECKLIST

Up-to-date data

Data should be up-to-date in order to obtain maximum value from the data analysis.

Missing values

Count missing values and analyze where in the data they are missing. Missing values can disrupt some analyses and skew the results.

Duplicates

Duplicate IDs indicate multiple records for one person, e.g. someone holds multiple functions at the same time.

Numerical outliers

Numerical outliers are fairly easy to detect and remove. Define minimum and maximum to spot outliers easily.

Check IDs

Check data labels of all the fields to see whether some categorical values are mislabeled.

Define valid output

Define valid data labels for categorical data. Define data ranges for numerical variables. Non-matching data is presumably wrong.

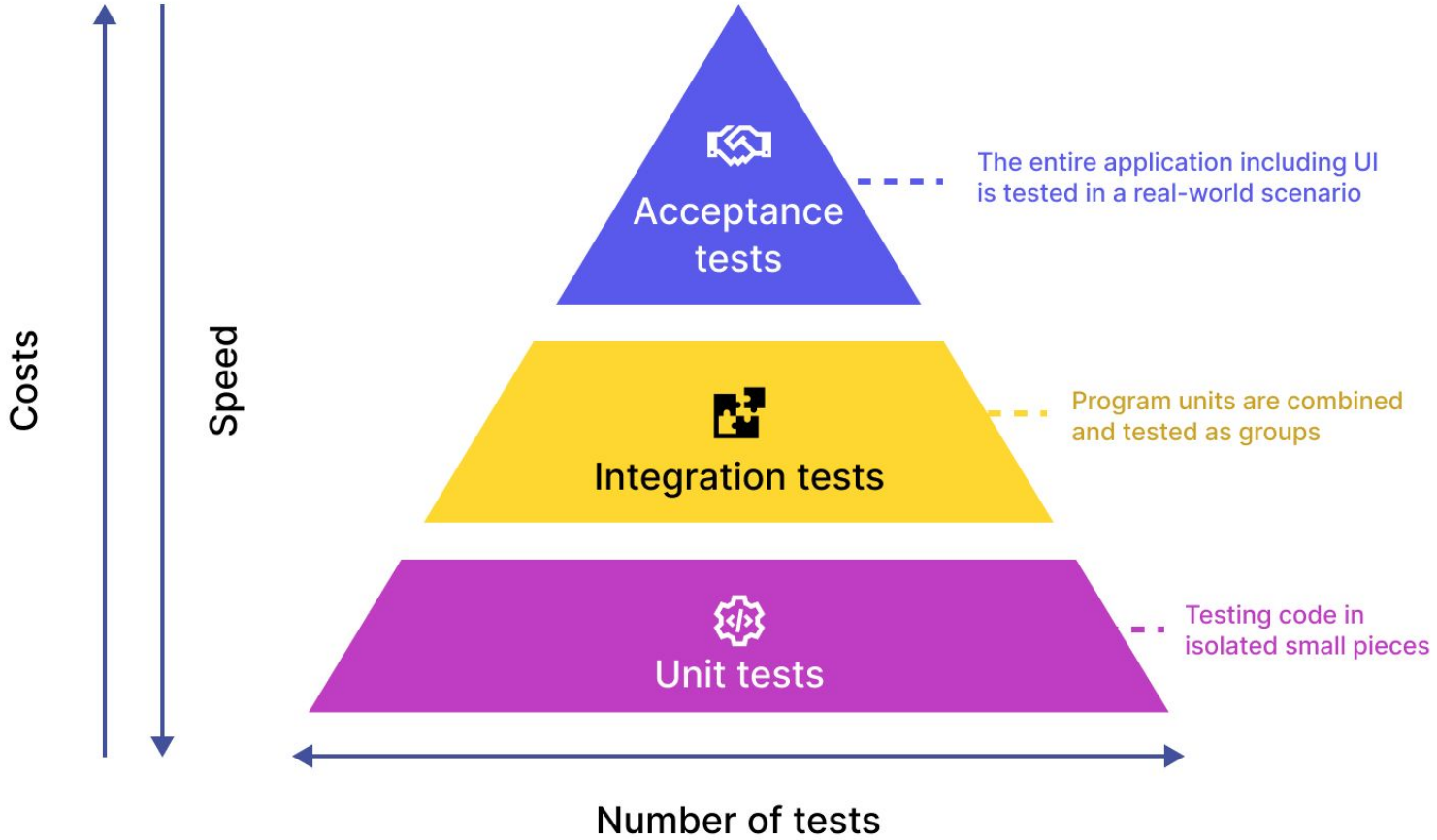
AIHR
BLOG & ACADEMY

DATA CLEANING CHECKLIST

- Duplicate Data
- Missing Values
- Inconsistent Units
- Outlier Detection
- Accuracy & Validity Check
- Data Type Validation
- Date and Time Formats
- Text and String Cleaning
- Consistency Checks

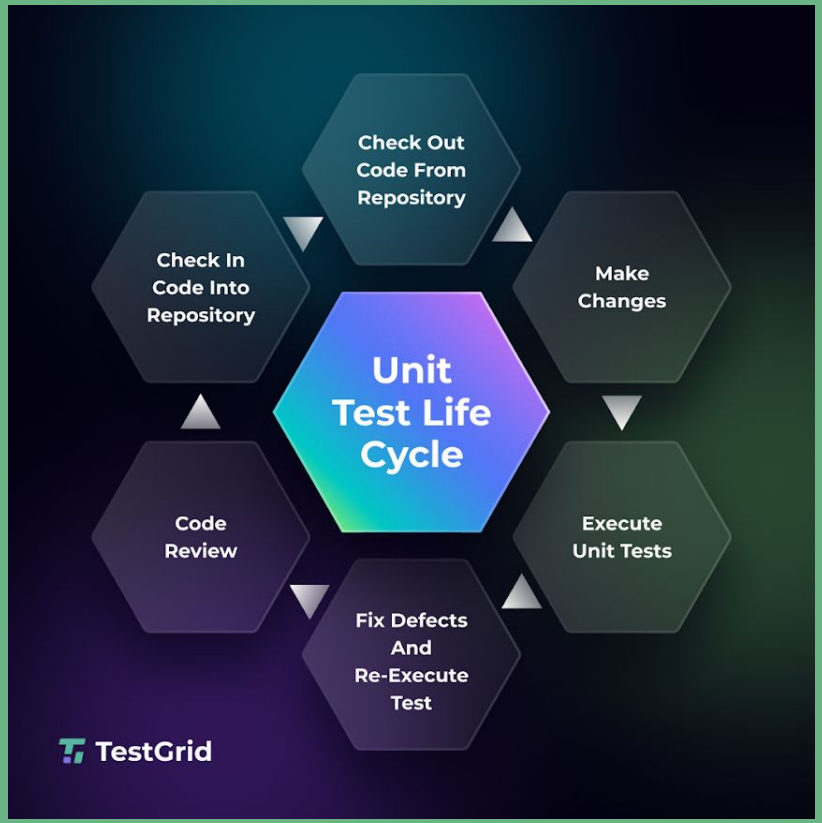
in VICTORY BROWN

Testing


Costs
Speed
Acceptance tests
The entire application including UI is tested in a real-world scenario
Integration tests
Program units are combined and tested as groups
Unit tests
Testing code in isolated small pieces
Number of tests

Unit Testing

- Unit testing is the practice of **testing small pieces of code** (like individual functions) to make sure they work correctly
- **Why it's important:** It helps catch bugs early, makes code safer to change, and improves reliability
- **Big picture:** Unit tests build confidence in your system by ensuring each building block works before everything is combined



Integration

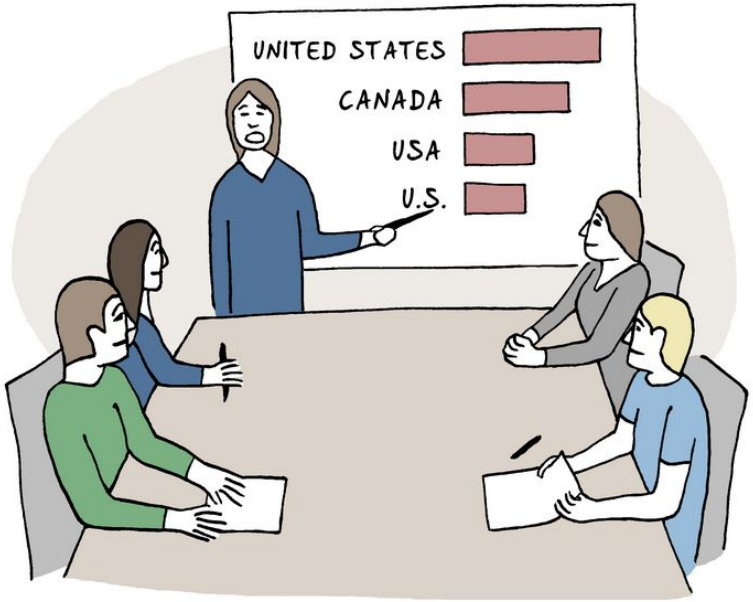
- Integration testing checks that **different parts of a program work correctly when combined** (e.g., frontend + backend, or multiple modules together)
- **Why it's important:** It helps catch issues that only appear when components interact, like data mismatches or communication errors
- **Big picture:** Integration tests make sure the system works as a whole, not just as isolated pieces

PyTest

- **pytest** is a Python testing framework used to write and run automated tests for your code
- **Why it's popular:** it has simple syntax, clear error messages, and powerful features (like fixtures and plugins) that make testing easier and more flexible

test_math_operations.py

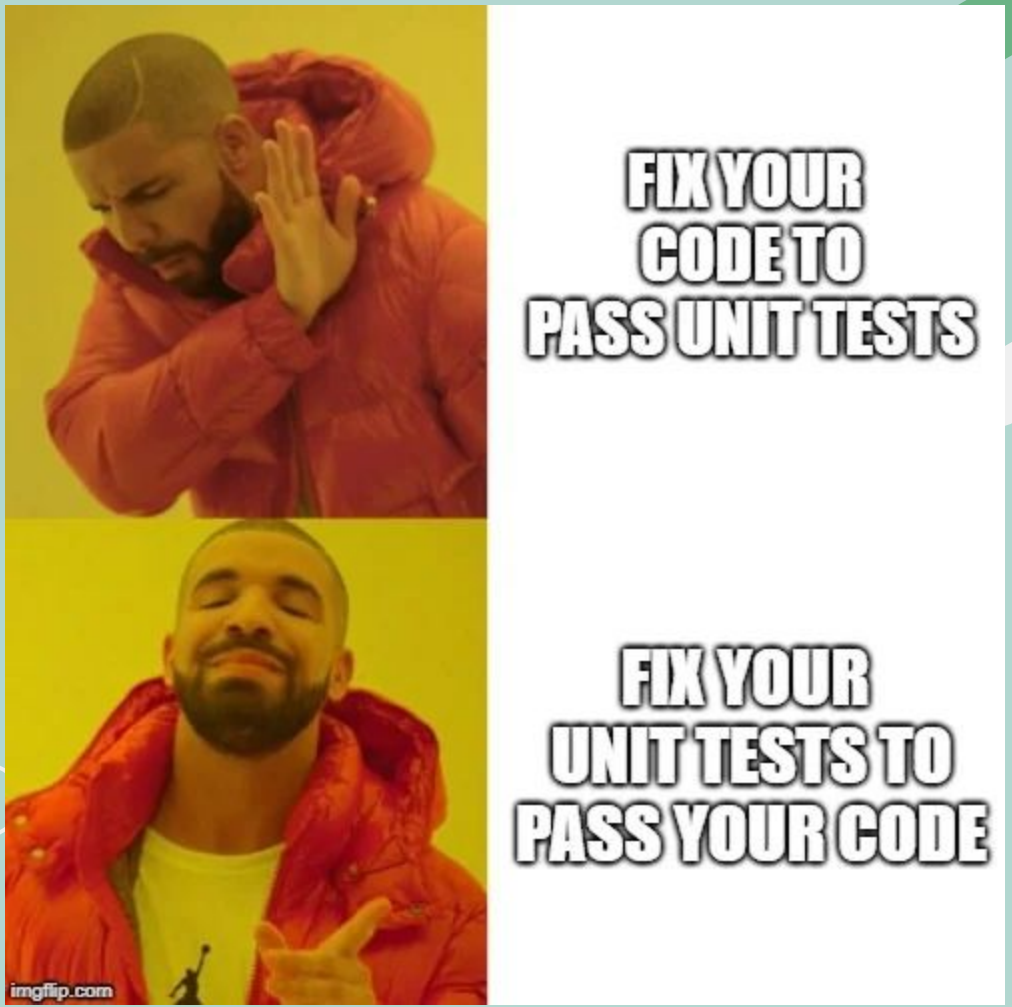
```python
import pytest
from math_operations import multiply

@pytest.mark.parametrize(
    "a, b, expected",
    [
        (2, 3, 6),  # 2 multiplied by 3 equals 6
        (4, 5, 20),  # and so on...
        (6, 7, 42),
        (0, 10, 0),
        (-1, -1, 1)
    ]
)
def test_multiply(a, b, expected):
    assert multiply(a, b) == expected
```



AS YOU CAN SEE, OUR TOP MARKETS ARE UNITED STATES, CANADA, USA AND THE U.S.

Dataedo /cartoon

Piotr@Dataedo


FIX YOUR CODE TO PASS UNIT TESTS
FIX YOUR UNIT TESTS TO PASS YOUR CODE
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TONIGHT WE TEST
IN PRODUCTION!!!

EXERCISE

Lfg